

ZigBee and Low-Power Wireless Networks: Architecture, Applications, and Challenges

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Abstract—Internet of Things (IoT) has seen tremendous growth in recent years; there is thus increased need for reliable and energy-efficient wireless connectivity. ZigBee has emerged as one of the most popular technologies for low-power wireless networks and operates on IEEE 802.15.4 standard. ZigBee is designed to facilitate low bit rates, long battery life, and secure communications in interconnected networks of devices. This study explores the key concepts of low-power wireless networking, analyzing the structure, protocol stacks, network topologies, and security features of ZigBee technology. Also highlighted in this paper is a use case for industrial smart factories, showcasing how ZigBee can be deployed in the real world. ZigBee technology is benchmarked against Bluetooth Low Energy (BLE) and LoRaWAN technologies. Analysis reveals ZigBee as a technology that provides a balanced solution in terms of energy consumption, reliability, scalability, and mesh network support, rendering it ideal as a communication standard for industrial IoT and wireless sensor networks.

Index Terms—ZigBee, IEEE 802.15.4, Wireless Sensor Networks, Internet of Things

I. INTRODUCTION

The increase in the usage of IoT has changed the way information is collected, processed and transmitted in the manufacturing industry, as well as the home of consumers. An IoT system is made up of multiple sensors, actuators, and embedded devices connected wirelessly but with the challenge of having to perform while being powered by extremely limited energy resources [1].

Common wireless technologies like Wi-Fi can support high data transmission speeds, but they consume more energy than necessary. Many industrial applications that involve monitoring and controlling processes need wireless communication technologies capable of working for long periods using minimal energy while transferring minimal data [2].

Low power wireless communication networks have been developed as a result of the need for such applications. In terms of wireless sensor networks and industrial IoT, some of the most prominent technologies include ZigBee. The basis of ZigBee is the IEEE 802.15.4 standard [3].

The main purpose of this paper is to explore ZigBee technology and low power wireless networks, describe their architecture and functioning, consider industrial application aspects, study strengths and weaknesses, and review future perspectives in this area.

II. FUNDAMENTALS OF LOW-POWER WIRELESS NETWORKS

A. Characteristics of Low-Power Wireless Networks

A low-power wireless network is a type of communication network that focuses on consuming minimal power and connecting devices reliably. This type of network is often used in wireless sensor networks (WSNs), home automation, industry automation, health care management, and environmental monitoring systems [4].

A low-power wireless network usually involves sensor nodes, wireless links, and a coordinator or gateway. The role of the sensor node is to gather environmental data and forward the collected information through wireless communication links to the central processing node. Because most of the sensor nodes use batteries for energy supply, conserving energy becomes crucial [1], [4].

- Low energy consumption
- Low implementation cost
- Small packet sizes
- Moderate communication range
- Support for mesh networking
- Long battery lifetime
- Scalability for large device deployments

B. Wireless Sensor Networks and Communication Technologies

Wireless sensor networks can be considered one of the most important applications of low power wireless communication techniques. As Akyildiz et al. have stated [1], "wireless sensor networks provide distributed sensing and monitoring by using a large number of inter-connected sensor nodes which collectively gather and relay information". This kind of network is extensively utilized in fields like environmental monitoring, industrial control, healthcare and smart cities since they help in collecting and transmitting information.

Among low-power communication technologies, there are ZigBee, Bluetooth Low Energy (BLE), LoRaWAN, Thread, and WirelessHART. The most promising technology is ZigBee due to its combination of low power consumption, ability to work in a mesh mode, and standardization of interoperability between devices regardless of their manufacturer. Moreover,

ZigBee networks provide scalability and robustness, which make this technology perfect for IoT and wireless sensor networks [2], [5].

III. ZIGBEE ARCHITECTURE AND IEEE 802.15.4

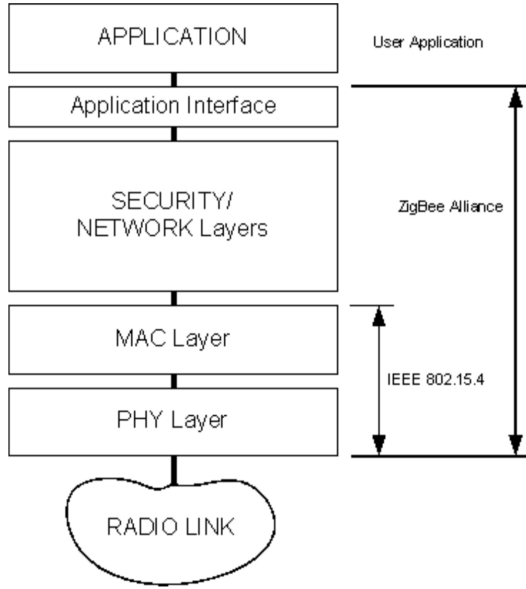


Fig. 1. Communication model adapted to a ZigBee stack. Adapted from AN3403 [6].

According to Fig. 1, the ZigBee protocol stack builds on the IEEE 802.15.4 physical and medium access control layers by providing additional features in the network layer, the security layer, and the application layer.

ZigBee is a low-power, low-data rate communication protocol that works through wireless transmission. This protocol is an extension of the IEEE 802.15.4, which includes the PHY layer and MAC layer functionalities. ZigBee goes further to provide functionalities like networking, routing, security, and application functions [3], [7].

The ZigBee protocol stack consists of four main layers:

- 1) Physical Layer (PHY)
- 2) Medium Access Control Layer (MAC)
- 3) Network Layer (NWK)
- 4) Application Layer (APL)

The Physical Layer is concerned with wireless data transmission and reception. IEEE 802.15.4 allows communication on several frequency bands such as 868 MHz, 915 MHz, and 2.4 GHz ISM band, which is available globally. The data rate is varied from 20 kbps to 250 kbps depending on the chosen frequency band [3].

MAC Layer is used to regulate access to wireless channels, as well as ensuring data integrity with carrier sense multiple access with collision avoidance (CSMA/CA), frame validation, acknowledgment, and synchronization capabilities [8].

The Network Layer helps establish and control ZigBee network. One of the most valuable properties of ZigBee is

that it offers mesh networking. The mesh topology allows forwarding of data packages via other intermediate devices; thus, the communication between devices is possible even if there is no connection directly between them [2].

The Application Layer consists of application dependent functions and profiles. The Application Layer is responsible for the compatibility of devices produced by various manufacturers by defining their standard behavior and communication protocols [7].

A. Network Topologies

The three types of network topologies in ZigBee are as follows:

Star Topology: All nodes have direct communications with the central coordinator. Star topology is easy to implement but has low coverage capability.

Tree Topology: Communications take place in hierarchical manner. Children nodes communicate through their parent nodes to cover greater network area without using too complex routing algorithms.

Mesh Topology: Nodes may have multiple routes for communications. Mesh networks offer fault tolerance since alternate paths exist in case of any breakage in communicational links [9]. As shown in Fig. 2, ZigBee supports multiple

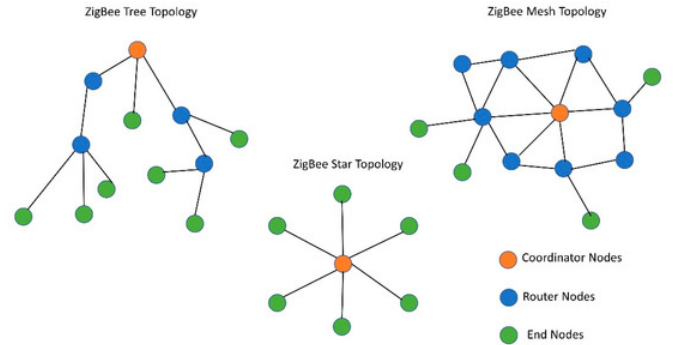


Fig. 2. Common ZigBee network topologies: star, tree, and mesh. Adapted from [10].

network topologies that provide different levels of coverage, scalability, and reliability.

B. Device Types

Three types of devices exist in the ZigBee network, namely:

ZigBee Coordinator (ZC): Coordinator initializes the network and manages the network traffic. There exists only one ZigBee Coordinator per ZigBee network.

ZigBee Router (ZR): ZigBee Routers are responsible for forwarding packets between other nodes.

ZigBee End Device (ZED): ZEDs are responsible for performing sensing and actuating tasks. They neither take part in packet forwarding nor routing and can sleep [4].

IV. INDUSTRIAL APPLICATION EXAMPLE: SMART FACTORY MONITORING

Smart factory monitoring is one of the major use cases of ZigBee technology. Today, any industrial environment needs

constant monitoring of machinery, facilities, and environmental factors to maintain high productivity and avoid any unexpected issues. Sensor networks that use ZigBee can offer a flexible and inexpensive method for gathering and exchanging this information [9].

With a smart factory, ZigBee sensors can be attached to machinery to monitor parameters like temperature, vibrations, humidity, pressure, and energy usage. The gathered information will then be transmitted over the ZigBee wireless sensor network to a gateway, from where it will be forwarded to another system for processing [4], [5].

The communication demands for an industrial monitoring system are as follows:

- Reliable data transmission
- Low power consumption
- Scalability for large numbers of devices
- Secure communication
- Low deployment cost
- Fault tolerance

These requirements are fulfilled by ZigBee due to its capacity for mesh networking, low power consumption, and ability to facilitate secure communication. Devices running on battery could last for years without maintenance, which makes ZigBee well-suited for industrial usage [9].

Moreover, ZigBee mesh networks feature a self-healing capability. If one route becomes unusable due to interference or malfunctioning equipment, other communication routes can be automatically chosen. Such capabilities make ZigBee networks more reliable, thus helping to maintain the functioning of the monitoring systems [2].

Other than condition monitoring, the ZigBee wireless sensor network can also facilitate predictive maintenance. This involves monitoring the machines continuously to predict any failures before they cause production downtime. The sensor data collected from the network can then be analyzed using data analytics tools to determine any unusual operation. With that, maintenance can be conducted preemptively rather than reactively. Thus, this would lead to improved production processes and reduced cost of production. In this way, ZigBee is suitable for Industrial Internet applications [5], [9].

V. EXPANDED APPLICATIONS: SMART AGRICULTURE AND UTILITIES

Apart from factory automation, ZigBee has been used widely in agriculture and utility management. Precision agriculture involves the use of wireless sensor network systems powered by ZigBee to measure soil moisture, soil temperature, humidity, and crop status in real-time. With such data, automatic irrigation and pest detection is possible, hence facilitating informed decision-making for improved productivity while minimizing energy use [11].

Use of current information provided by environmental sensors instead of programmed times enables ZigBee-powered agricultural operations to be sustainable and efficient [12].

In smart cities and utilities, ZigBee is common in several applications including intelligent street lighting. ZigBee-powered

light posts interact with the control center to minimize energy consumption and maintenance tasks [13].

Moreover, Zigbee is becoming a popular choice for use in smart power transmission and distribution, as it allows constant monitoring of network parameters, faults detection, and interaction with energy management systems. Such solutions ensure the high performance, safety, and sustainability of today's utility grids and promote the overall development of smart city systems [14].

VI. ADVANTAGES AND LIMITATIONS OF ZIGBEE

Similar to any communication technology, there are many advantages and limitations associated with ZigBee. It is imperative that one should understand these advantages and limitations before selecting the right communication standard for the particular application.

A. Advantages

The main advantages of ZigBee include:

- Very low power consumption
- Long battery lifetime
- Low implementation cost
- Support for mesh networking
- Scalability for large networks
- Reliable communication through self-healing mechanisms
- Standardized interoperability between devices

The most important advantages include efficiency in terms of energy usage. With a single battery ZigBee-based products can work for many years, which makes them perfect for wireless sensing networks [2], [4].

Another major benefit is mesh networking. Data transmission becomes possible through different paths within the network, thus providing greater coverage than point-to-point communications systems [7].

B. Limitations

While ZigBee has many advantages, it also suffers from some limitations:

- Limited data rate
- Short communication range
- Potential interference in the 2.4 GHz band
- Not suitable for high-bandwidth applications
- Higher latency compared to some wired industrial networks

The maximum rate of ZigBee transmission speed is 250 kbps, which is sufficient to transmit data but is not enough when dealing with voice data, video data, or any large amount of data transmission. Also, there can be interferences from other networks using the 2.4 GHz frequency band [7].

VII. SECURITY CONSIDERATIONS

Data security is one of the most crucial aspects that should be considered for reliable transmission in wireless communication systems. It becomes especially important for the IoT industry and in applications when some sensitive data can be

transmitted via wireless channels. As ZigBee devices can be utilized in smart homes, in various health systems, and other purposes, there should be provided adequate security measures to guarantee data confidentiality, integrity, and availability [3], [7].

There are multiple security features used in ZigBee, and all of them provide the possibility to make the process of exchanging information more secure. For example, one of the main measures used by ZigBee is the usage of the Advanced Encryption Standard (AES) with a key length equal to 128 bits or AES-128. This measure allows making sure that transmitted information cannot be interpreted by unauthorized third parties [3].

Moreover, there is the opportunity to employ different authentication mechanisms and ensure the authenticity of all parties involved in communication. Besides, there are mechanisms to check integrity and prevent any alteration of transmitted information that is quite important in some cases [7].

The ZigBee network comprises a trust center that handles all security keys distribution and controls access to the devices in a network. Trust center manages the keys, authenticates devices, and secures a network. Security management is simplified through centralization of security management in the trust center [7].

However, these safety measures do not make ZigBee networks impervious to attacks. These attacks can range from listening attack, replay attack, denial-of-service, to attacks on encryption keys among others. Hackers will seek to listen to wireless transmission, disrupt links, and take advantage of any possible weakness in the configuration. Furthermore, outdated device firmware or poor key management will also contribute to increased vulnerability of the network [4].

Organizations can increase their security by ensuring effective implementation of key management processes, updating firmware of the devices, restricting access to the physical location of network devices, and continuous monitoring of network operations for any unusual behavior [7].

One more aspect worth considering is the growing use of ZigBee devices alongside with cloud IoT platforms and smart building systems. As the number of connected devices increases, the risk of compromising the network increases as well. Thus, apart from the wireless communication, the security must also include gateways, cloud-based solutions, and other elements used in the process of managing connected devices. Combining different techniques such as encryption, access control, intrusion detection, and software updates is the key to ensuring better resilience of ZigBee systems against new types of cyber attacks [4], [7].

In summary, ZigBee offers a set of tools required for achieving strong protection of WSNs. Although there still exist security issues in terms of ZigBee technology, they can be addressed by implementing appropriate strategies [3], [7].

VIII. COMPARISON WITH BLE AND LORAWAN

There are various types of wireless communication networks used in IoT and industrial solutions. The most popular alternatives to ZigBee are BLE and LoRaWAN. Every technology has its unique characteristics and features making it applicable only to specific use cases [5].

A. Bluetooth Low Energy (BLE)

BLE stands for Bluetooth Low Energy and is a type of Bluetooth which uses less energy than traditional Bluetooth technologies. The BLE is usually used in wearable technologies, smartphones, fitness wearables, and various medical applications. BLE provides high-speed communication and consumes less power than other technologies. Besides, this technology is easy to integrate with new smartphones. However, it is characterized by short-range radio communication, thus unable to create the large network [5].

B. LoRaWAN

LoRaWAN stands for Long Range Wide Area Network and is designed for long-range data transfer and consumes little power. Therefore, this technology is the most suitable for smart farming, environment monitoring, and smart city management. However, LoRaWAN has relatively low transmission speed and high latency compared to ZigBee and other low-power technologies [5].

C. ZigBee

ZigBee represents a balanced solution combining the advantages of BLE and LoRaWAN. Thus, it has an average range, consumes little power and may be used as a mesh network. Mesh network implies transferring data from one node to another [2], [7].

TABLE I
COMPARISON OF ZIGBEE, BLE, AND LORAWAN

Feature	ZigBee	BLE	LoRaWAN
Range	Medium	Short	Long
Power use	Low	Very low	Very low
Data rate	Medium	High	Low
Mesh network	Yes	Limited	No
Industrial IoT	High	Medium	High
Smartphone support	Limited	High	No
Battery lifetime	Long	Long	Very long

The analysis in Table I shows that there is no such communication technology that is ideal for all possible usage cases. BLE technology can be considered effective for short range applications and personal devices because of compatibility with smartphones. LoRaWAN is highly effective at long range communications with extremely low power consumption. In addition, this technology is useful for tracking purposes and communication on rural areas. The combination of low energy consumption, reliability, and mesh networking makes ZigBee technology ideal for factories, smart buildings, and sensor networks.

It is important to consider your requirements when choosing the appropriate technology. For example, BLE technology can

be used for personal devices and short-range communication. On the contrary, LoRaWAN technology is better if communication happens at long range but with minimal amount of information transfer. Finally, ZigBee technology can be used for manufacturing, smart building, and sensor networks.

IX. ENERGY MANAGEMENT AND HARVESTING TECHNIQUES

However, despite being a low-power communication protocol, energy management still remains a critical problem in large-scale implementations of WSN using ZigBee-based sensors. In many cases, such as environment monitoring, intelligent agriculture and automation, sensors can be placed in remote and hard to access places. In such cases, frequent battery replacements can increase expenses on maintenance and make a system less sustainable overall. To overcome this problem, the idea of adding EH techniques to ZigBee-based sensor networks has been widely investigated in recent researches [4], [15].

The term energy harvesting implies the capturing of energy from surrounding environment and transformation to electricity that is used to supply sensor node with energy. One of the most commonly used energy harvesting techniques is solar EH. ZigBee-based sensors can use solar cells in conjunction with MPPT circuits in order to achieve maximum energy conversion efficiency. Such technique is appropriate in outdoor applications, such as intelligent agriculture, environment monitoring, SHM etc [15].

Another viable approach in consideration is the use of radio frequency (RF) energy harvesting. In this case, the energy will be sourced from electromagnetic fields present in the surroundings, which arise due to wireless communication networks. Even though the amount of energy is not abundant, RF energy harvesting will give a secondary source of energy that can sustain ultra-low-power ZigBee sensors in dense networks [16].

One more potential candidate is mechanical vibration energy harvesting. Mechanical vibrations from machinery, motors, and industrial equipment can produce electrical energy by harvesting either piezoelectrically or electromagnetically. The generated electrical energy will then help in powering wireless monitoring systems, thus avoiding battery replacement issues [17].

Combining energy harvesting systems with the ZigBee communication network provides various benefits such as increased life span of devices, decreased cost of maintenance, and greater environmental sustainability. In addition, energy harvesting can minimize the reliance on batteries and enable the deployment of self-powered wireless sensors that are capable of performing tasks over a prolonged period of time without requiring any human input. Therefore, energy harvesting will be vital in the future use of IoT based on ZigBee networks [15]–[17].

X. ADVANCED CHALLENGES: INTERFERENCE AND SCALABILITY

However, despite all of its benefits, the implementation of ZigBee in a large-scale environment faces some challenges. First of all, it is worth mentioning interference, which occurs at the 2.4 GHz frequency band shared with other communication standards like Wi-Fi and Bluetooth. In densely-populated areas, such as smart buildings and factories, co-existence with other technologies can be disadvantageous in terms of increased packet collisions and reduced reliability [18]. Several solutions to this problem include channel selection algorithms and better signal propagation mechanisms in ZigBee networks [18].

Another challenge to be taken into consideration is scalability, which implies additional overhead in terms of routing and network management and leads to increased energy consumption when applied in large-scale networks. Router nodes suffer from energy depletion, as they forward data packets from one device to another. This results in a shorter lifetime of router nodes in large-scale applications and, thus, decreases network performance. Hence, proper network planning, as well as energy harvesting, can ensure good performance of large-scale ZigBee networks [19].

XI. FUTURE TRENDS

As far as the future of ZigBee is concerned, its future is strongly related to the evolution of the Internet of Things (IoT), the Industrial Revolution 4.0, and smart infrastructure. Nowadays, there is a great interest in the integration of ZigBee systems with cloud computing platforms, edge computing platforms, and building automation systems. Furthermore, innovative projects like Matter attempt to enhance the compatibility between devices produced by different vendors, thus allowing ZigBee solutions to become a part of heterogeneous IoT systems [7].

Despite all the above, the popularity of ZigBee technology is threatened by competing solutions like Thread, Bluetooth Mesh, or even LoRaWAN. Nonetheless, the wide adoption of ZigBee networks is justified by their robustness and reliability, the availability of a mature ecosystem, as well as low energy consumption. In the foreseeable future, innovations related to ZigBee technology can be expected in the field of energy-efficient operation, secure communication, as well as interoperability with other IoT technologies [5], [7].

XII. CONCLUSION

The fast growth of the Internet of Things (IoT) is associated with an increased need for wireless communication solutions capable of offering reliable data transmission at a reduced energy cost. Among those, ZigBee has been established as one of the key wireless communication solutions as a result of its ability to operate in such fields as wireless sensor networks, industrial automation, and smart buildings.

This paper reviewed the core characteristics of low-power wireless networks along with providing information about the protocol stack of ZigBee and its operation based on IEEE

802.15.4. In addition to this, an overview of potential use cases of ZigBee in industrial applications, security options, methods for energy harvesting, and trends related to future developments was provided.

According to the analysis, ZigBee has a number of benefits, such as low power consumption, high battery life, mesh networking capability, scalability, and reliability. The combination of those features makes ZigBee appropriate for applications in industrial Internet-of-Things and wireless sensor networks. However, there are some disadvantages associated with the use of the technology, namely, low data transfer rate and communication distance.

In particular, comparing ZigBee to BLE and LoRaWAN shows that while BLE is an ideal technology for devices used by individuals and operates effectively within short distances, LoRaWAN is great for monitoring long distances; however, ZigBee is a more balanced option that is characterized by energy efficiency, reliability, security, and mesh networking.

In conclusion, from this research paper, it is clear that ZigBee technology has continued to be an excellent mode of communication in wireless communication networks. The fact that it is energy-efficient, scalable, reliable, and safe means that it can offer an efficient communication tool in many industries and in the Internet of Things. Additionally, advances that are being made concerning energy harvesting, interoperability, and enhanced security will continue to play an essential role in its adoption in future smart environments. In spite of increasing competition from new technology, ZigBee will remain a mature communication technology.

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